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(a) Observer moves towards Source
(b) Observer receding away from the Source

(iii) Both are in motion
(a) Source and Observer approach each other
(b) Source and Observer recede from each other
(c) Source chases Observer
(d) Observer chases Source

***Stationary observer* and *stationary source* means the observer and source are both at rest with respect to medium respectively**

11.11.1 Source in motion and the observer at rest

(a) Source moves towards the observer

Suppose a source S moves to the right (as shown in Figure 11.46) with a velocity v_s and let the frequency of the sound waves produced by the source be f_s. We assume the velocity of sound in a medium is v. The compression (sound wave front) produced by the source S at three successive instants of time are shown in the Figure 11.46. When S is at position x_1 the compression is at C_1. When S is at position x_2, the compression is at C_2 and similarly for x_3 and C_3. Assume that if C_1 reaches the observer's position A then at that instant C_2 reaches the point B and C_3 reaches the point C as shown in the Figure 11.46. It is obvious to see that the distance between compressions C_2 and C_3 is shorter than distance between C_1 and C_2. This means the wavelength decreases when the source S moves towards the observer O (since sound travels longitudinally and wavelength is the distance between two consecutive compressions). But frequency is inversely related to wavelength and therefore, frequency increases.

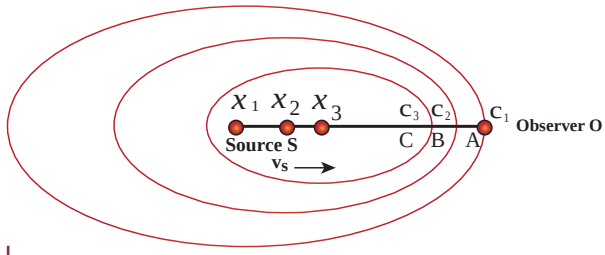


Figure 11.46 Source S moves towards an Observer O (right) with velocity

Let λ be the wavelength of the source S as measured by the observer when S is at position x_1 and λ' be wavelength of the source observed by the observer when S moves to position x_2. Then the change in wavelength is $\Delta\lambda = \lambda - \lambda' = v_s t$, where t is the time taken by the source to travel between x_1 and x_2. Therefore,

$$\lambda' = \lambda - v_s t \qquad (11.83)$$

$$\text{But} \quad t = \frac{\lambda}{v} \qquad (11.84)$$

On substituting equation (11.84) in equation (11.83), we get

$$\lambda' = \lambda \left(1 - \frac{v_s}{v}\right)$$

Since frequency is inversely proportional to wavelength, we have

$$f' = \frac{v_s}{\lambda'} \quad \text{and} \quad f = \frac{v_s}{\lambda}$$

$$\text{Hence,} \quad f' = \frac{f}{\left(1 - \frac{v_s}{v}\right)} \qquad (11.85)$$

Since, $\frac{v_s}{v} \ll 1$, we use the binomial expansion and retaining only first order in $\frac{v_s}{v}$, we get

$$f' = f\left(1 + \frac{v_s}{v}\right)\mathrm{v} \qquad (11.86)$$

(b) Source moves away from the observer:

Since the velocity here of the source is opposite in direction when compared to case (a), therefore, changing the sign of the velocity of the source in the above case i.e, by substituting ($v_s \rightarrow -v_s$) in equation (11.83), we get